



Design of metallization layers for PZT piezoelectric ceramics and joining to 4J29 Kovar alloy



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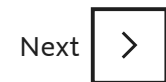
Highlights

- The addition of Ti layer to the metallization layers improves the shear strength of joint.
- The Sn-3.0Ag-0.5Cu solder exhibits excellent wetting on the (Ti + Ni + Au)metallization layers.

- An optimal thickness of the Au layer proves advantageous for solder wetting on the metallization layers.
- The joint's maximum shear strength is 29.0 ± 1.4 MPa at layer thicknesses of 150 nm (Ti), 1000 nm (Ni), and 300 nm (Au).

Abstract

To enhance the bonding strength of PZT ceramics/4J29 Kovar alloy joints, the metallization layers on PZT ceramics were strategically designed, fabricated, and optimized. Subsequently, the metallized PZT ceramics were successfully joined with the 4J29 Kovar alloy using Sn-3.0Ag-0.5Cu (SAC305) solder. In this study, the wetting behavior, joining mechanism and the effect of metallization layers thickness on the joining strength were investigated. The (Ti + Ni + Au) multilayer structure enhances wetting, as Au dissolves into the solder to improve wetting, while Ni serves as a barrier to prevent direct contact between the solder and Ti, thereby ensuring the integrity of the metallization layer. The joining mechanism for PZT ceramics/solder and solder/4J29 Kovar alloy involves the reaction between Ni (in the metallization layer) and Sn (in the solder), and Sn diffusion into 4J29 Kovar alloy, respectively. The highest joint strength of 29.0 ± 1.4 MPa was achieved with (Ti + Ni + Au) metallization layers (Ti:100 nm, Ni:1000 nm, Au:300 nm), marking a 120% improvement over the adhesive currently used for joining PZT ceramics to 4J29 Kovar alloy. Our work presents a promising method for joining PZT ceramics with 4J29 Kovar alloy, which is applied in transducers as a core component for oil logging.



Keywords

PZT ceramics; Metallization; Reflow soldering; Shear strength; Wettability

1. Introduction

The ever-rising demand for the better properties used in piezoelectric devices such as actuators, sensors, and transducers [1] require the better joining of ceramics and metals for the structure(metals)-function(ceramics) integration. However, due to the mismatch in coefficients of thermal expansion (CTE) and differences in physical and chemical properties, the joining of ceramics to metals poses a challenging issue [2]. $\text{Pb}(\text{Zr}_{1-x}\text{Ti}_x)\text{O}_3$ (PZT) based ceramics are consistently selected as the functional component for piezoelectric devices such as transducers applied in the oil logging [3] due to their outstanding piezoelectric properties, high Curie temperature (T_c), elevated mechanical quality factor, and commendable electromechanical coupling coefficient [4], [5]. When considering the structural component for piezoelectric transducers, the 4J29 Kovar alloy stands out as an excellent choice due to its low coefficient of thermal expansion (CTE) [6], which closely matches that of PZT. Nevertheless, the organic adhesive system employed for bonding PZT ceramics and 4J29 Kovar alloy (Fe-29Ni-17Co) in piezoelectric transducers presents drawbacks such as aging at elevated temperatures, low thermal conductivity, and insufficient joining strength [7].

To improve joining strength and operational temperature range of PZT ceramics/4J29 Kovar alloy piezoelectric transducers, developing a novel joining technique is necessary. However, limited by the curie temperature (300 °C) [8] of the PZT ceramics, traditional brazing [9], [10], [11], glass joining [12], diffusion bonding [13] are unsuitable due to their high joining temperature (normally above 400 °C). In contrast, soldering materials such as Sn-Pb [14], Sn-Zn [15], Sn-In [16], Sn-Bi [17], and Sn-Ag-Cu [18] offer a significant advantage with their lower joining temperatures below 300 °C [19]. Among the solder options mentioned above, Sn-3.0Ag-0.5Cu emerges as a highly promising alternative to organic adhesives, owing to its outstanding soldering performance, elevated strength, and superior thermal fatigue reliability. [20]. Employing Sn-3.0Ag-0.5Cu solder could potentially improve joint strength and prolong the operational lifespan of piezoelectric transducers.

The primary challenge in soldering PZT ceramics lies in the inadequate wettability of the solder on the ceramic surfaces, due to issues of chemical compatibility. [21]. One way to overcome this problem is to activate the solder with rare earth elements or active elements like Ti. Roman et al. [22] used Sn_2La solder to join Al_2O_3 and SiC ceramics and got the joints with shear strength of 7.5 MPa and 13.5 MPa, respectively. Wu et al. [23] joined SiC ceramics using Sn-Ag-Ti solder under the action of ultrasounds, and the shear strength could reach 60 MPa. However, these types of solders present challenges in terms of preservation and preparation, as both rare earth elements and titanium are prone to rapid oxidation. [24]. Another strategy to enhance solder wettability on ceramics involves the metallization of the ceramic surface. Compared to alternative metallization techniques, such as chemical plating [25] and cold spraying [26], magnetron sputtering, a form of physical vapor deposition [27], [28], offers

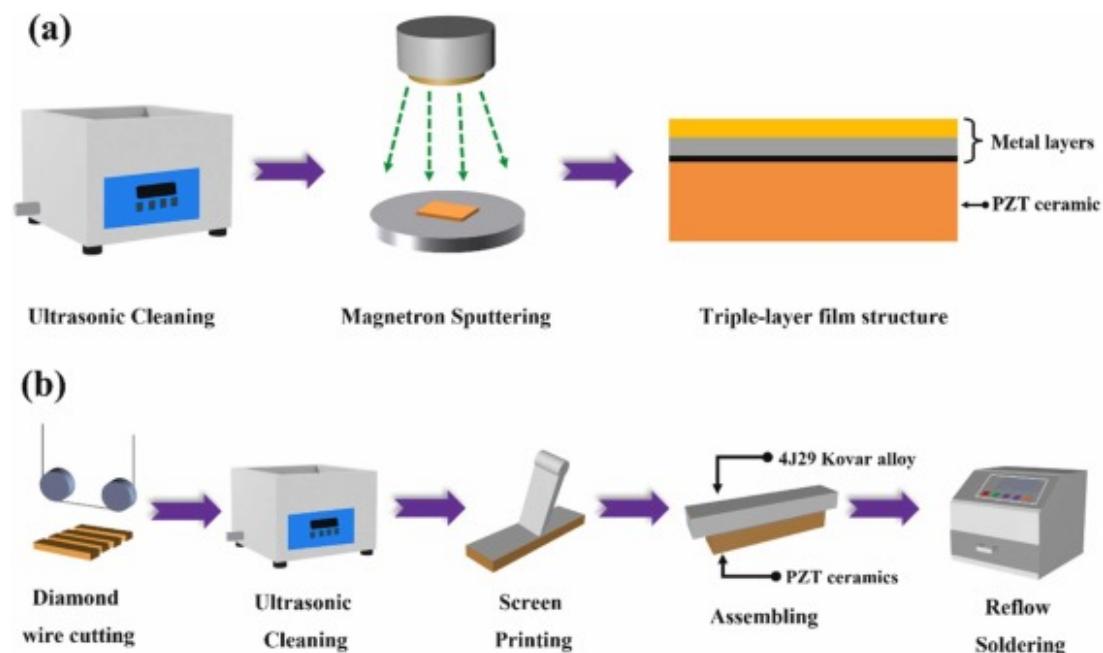
benefits including lower processing temperatures, higher sputtering rates, robust bonding strength, and eco-friendly attributes. Occasionally, a single metallization layer may not adequately facilitate ceramic bonding. Thus, applying multiple metal layers through sputtering becomes essential. Xia et al. [29] employed magnetron sputtering to deposit multi-layer metallic coatings on PZT ceramics and TC4 alloy, successfully achieving PZT/TC4 joints with a shear strength of 18.0 MPa using a Sn-3.0Ag-0.5Cu sheet.

Consequently, we propose the idea of assembling the piezoelectric device components (PZT ceramics and 4J29 Kovar alloy) using SAC305 solder. In the present work, to enhance the wettability of SAC305 solder on PZT ceramics, a multi-layer (Ti + Ni + Au) was designed and prepared using magnetron sputtering technology for the metallization of PZT ceramics. Subsequently, the joining of the 4J29 Kovar alloy to the metallized PZT using SAC305 solder was performed through reflow soldering. The wetting behavior of SAC305 solder on the PZT ceramics with different structures metallization layers were investigated. The interfacial microstructure of 4J29 Kovar alloy/SAC305 solder/metallization layers/PZT ceramics was also investigated using SEM, EDS and XRD. The wetting mechanism of the solder on the metallization layer were explored. Investigations were also conducted into how the strength of the joint is impacted by the thickness of the intermediate Ni layer and the surface Au layer in the study.

2. Experimental procedure

2.1. Materials preparation

In this research, the piezoelectric ceramics substrate employed was PZT-5 ceramics, which has a Curie temperature of 300°C. The chosen counterpart material was the 4J29 Kovar alloy, consisting of Fe-29Ni-17Co. For bonding the metallized PZT ceramics to the 4J29 Kovar alloy, SAC305 solder pastes, which include no-clean flux and an Sn-3.0Ag-0.5Cu alloy (provided by Senju Metal Industry Co.), were utilized. [Fig. 1\(a\)](#) depicts the procedure of metallization treatment employing Magnetron sputtering. Prior to metallization, the PZT ceramics were machined into the size of 20 mm × 5 mm × 3 mm by diamond wire saw. Then, these specimens were ultrasonically cleaned in anhydrous ethanol and deionized water for 15 mins, respectively. Following the cleaning process, the samples were dried and put into the magnetron sputtering chamber to finish the metallization process. It was noteworthy that the magnetron sputtering targets of Ti, Ni, and Au possessed a purity level equal to or exceeding 99.99%. In the experiment, the variation of the thickness of Ni and Au layers was controlled by the magnetron sputtering time.



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Fig. 1. The fabrication process of the 4J29 Kovar alloy/SAC305 solder/PZT joints by reflows soldering process. (a) metallization layers preparation; (b) joining process.

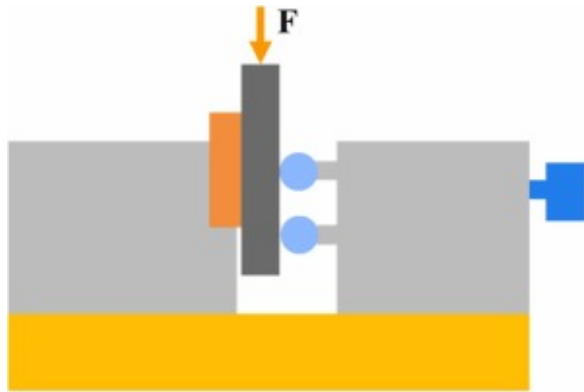
The subsequent joining process, as shown in Fig. 1(b), involved several precise stages. Before initiating the reflow soldering process, the 4J29 Kovar alloy was initially shaped into dimensions of 25 mm × 5 mm × 1.5 mm using electrical discharge machining, followed by an ultrasonic cleaning in anhydrous ethanol and deionized water, each for a duration of 15 minutes. Subsequently, SAC305 solder pastes were uniformly applied to the surface of metallized PZT ceramics via screen printing. The prepared PZT ceramics were then assembled with the 4J29 Kovar alloy, creating a sandwich configuration consisting of PZT ceramics/SAC305 solder/4J29 Kovar alloy. The assemblage was heated in a reflow oven with a processing parameter of 255 °C/15 seconds dwelling time.

2.2. Characterization of the samples

The interfacial microstructure of the joints was characterized by scanning electron microscopy (SEM, Magellan 400, FEI Co., U.S.A), equipped with an energy dispersive spectroscopy (EDS) for composition analysis. Melting point of the SAC305 solder alloy was analyzed by differential scanning calorimetry (DSC, DSC 3500, NETZSCH, Germany). The sessile drop method was employed to evaluate wetting behaviors of the solder on the metallized ceramics. Phase identification of the substrate materials (PZT ceramics and 4J29 Kovar alloy) and compositional analysis of the joints were conducted using X-ray diffraction (XRD, D8 Advance, Bruker AXS GMBH, Germany). The joining strength was evaluated using a 5566 universal testing machine in conjunction with a custom-made shear die, as depicted in Fig. 2, with a strain rate set to 1 mm/min. The shear strength can be calculated from the following Eq. (1):

$$S = \frac{F}{A} \quad (1)$$

Where S is the shear strength, MPa; F is the pressure applied by the universal testing machine, N; and A is the contact area of the connection between the PZT ceramics and the 4J29 Kovar alloy, m².



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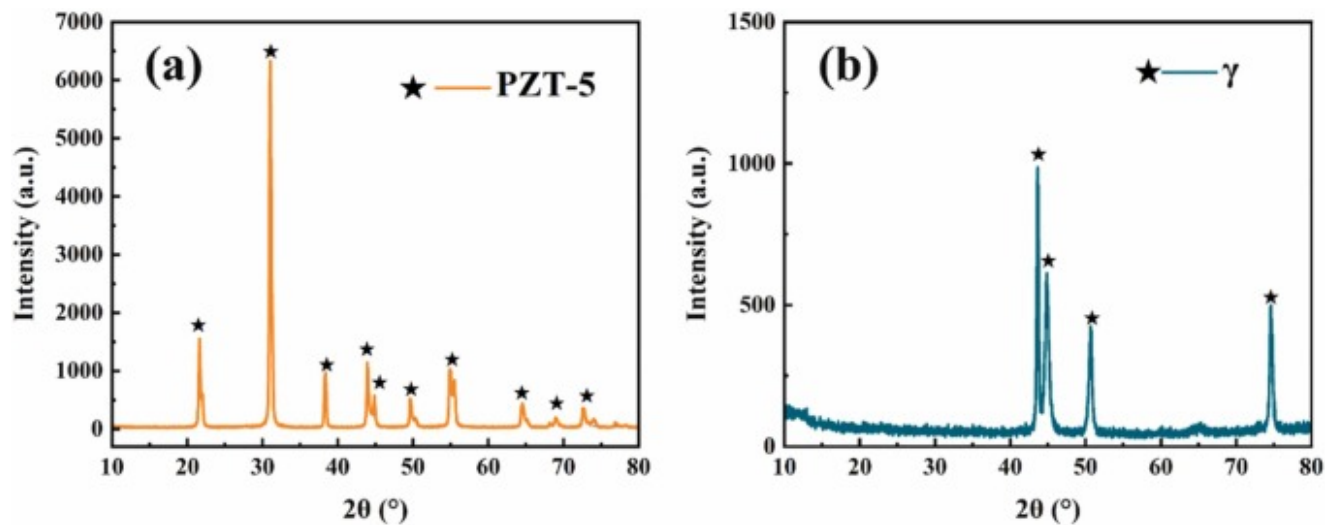
Fig. 2. Schematic diagram of shear strength test.

3. Results and discussions

3.1. Substrate characterization and design of metallization layers for PZT ceramics

3.1.1. Substrate characterization

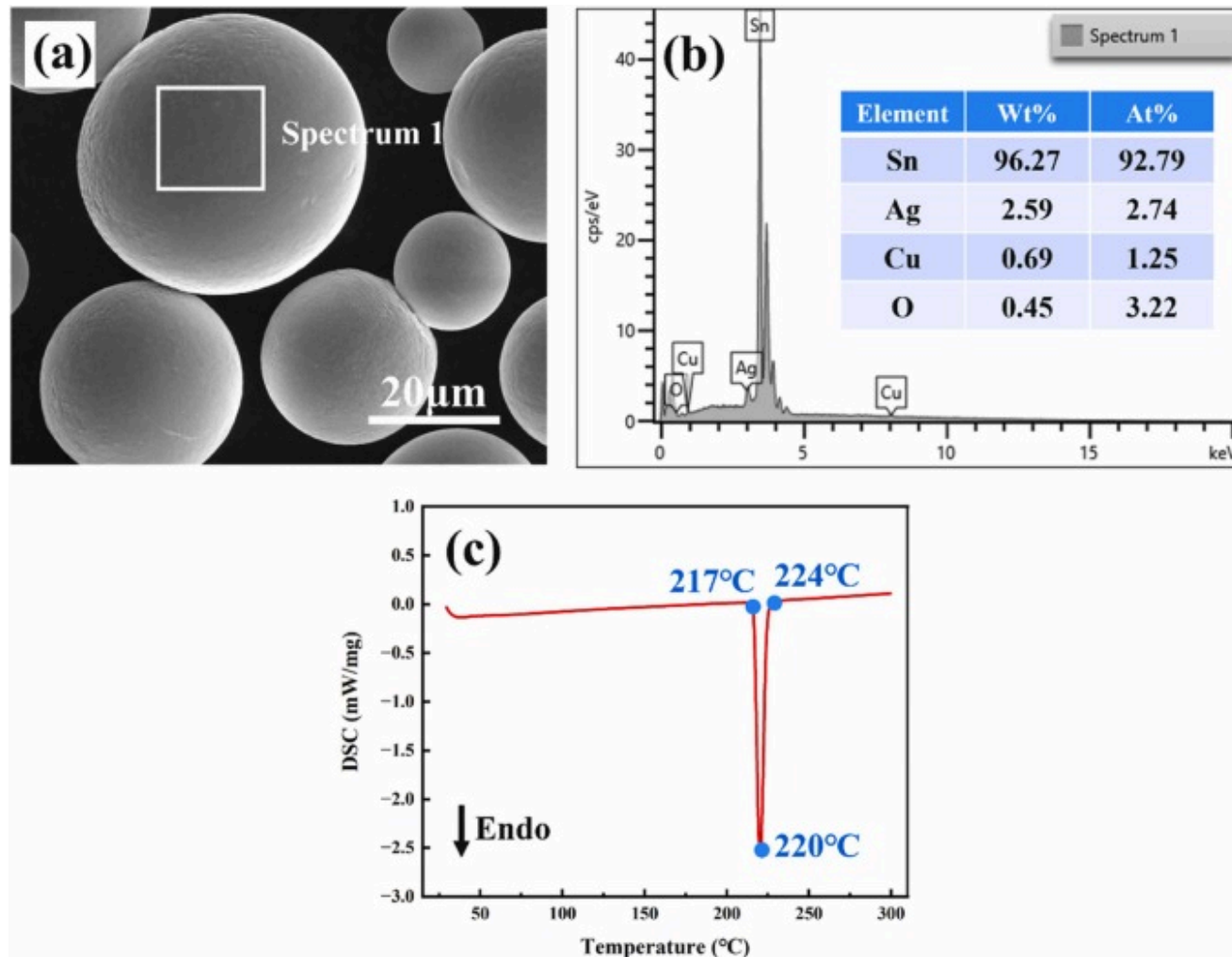
XRD patterns of PZT and 4J29 Kovar alloy are illustrated in Fig. 3. PZT ceramics was pure phase and 4J29 Kovar alloy was mainly composed of single-phasic austenite [30] (γ phase, face-centered cubic structure). Fig. 4(a) presents a SEM image of the SAC305 alloy powder, highlighting the alloy's outstanding sphericity and a particle size distribution ranging from 20 μm to 45 μm in diameter. Fig. 4(b) shows EDS results of the alloy powder, confirming that small amount of oxygen existed in the alloy powder, which was beneficial to reflow soldering. Fig. 4(c) shows the DSC curve of the SAC305 solder alloy powder. A heat absorption trough was observed at temperatures ranging from 217 $^{\circ}\text{C}$ to 224 $^{\circ}\text{C}$, with the lowest point of the trough at 220 $^{\circ}\text{C}$, suggesting that the solder's melting point lies within this range. Typically, the reflow soldering temperature is set 30 $^{\circ}\text{C}$ higher than the melting point of the solder alloy; hence, in this study, the reflow soldering temperature was chosen to be 255 $^{\circ}\text{C}$.



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Fig. 3. XRD pattern of (a) PZT; (b) 4J29 Kovar alloy.



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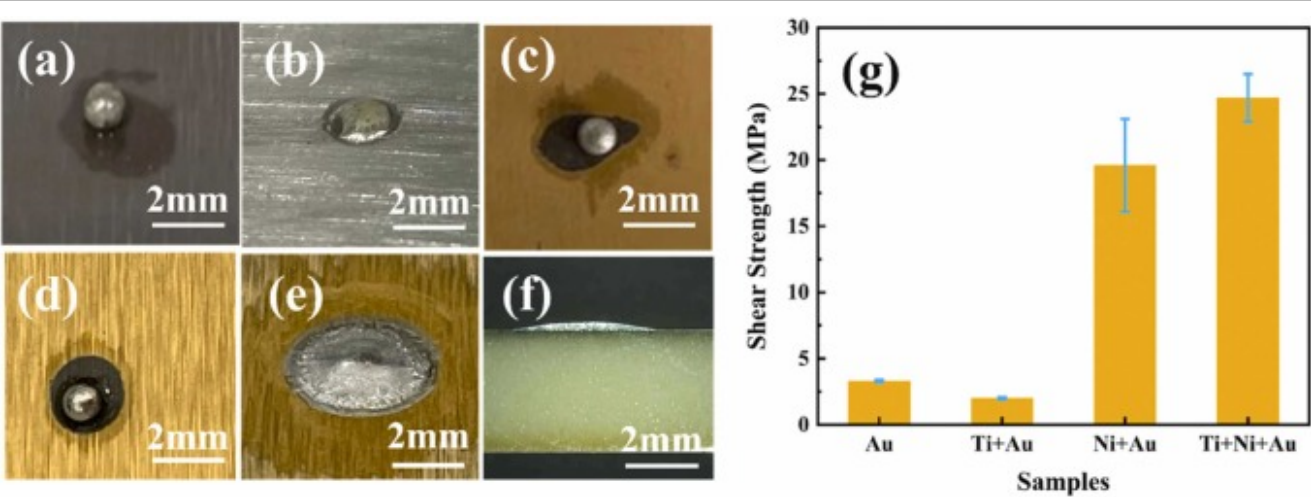
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Fig. 4. (a)SEM image; (b) EDS; (c)DSC curve of SAC305 solder alloy powder.

3.1.2. Design of metallization layers for PZT ceramics

The wettability of the SAC305 solder on the substrate plays a crucial role in determining the strength of the joints. Insufficient wetting on the substrate can result in the development of substantial voids within the joints, which markedly impacts the overall

joint strength. Fig. 5(a) and (b) show the wetting stage of SAC305 solder on the as-received PZT surface and 4J29 Kovar alloy surface, respectively. On the PZT ceramics, the solder presented ball shape, indicating that the solder almost did not wet the pure PZT ceramics. In contrast, it exhibits good wettability on the 4J29 Kovar alloy, as seen in Fig. 5(b). Therefore, metallization of the PZT ceramic was essential to obtain robust joints.



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Fig. 5. Wetting state of SAC305 solder on different metallization layers surfaces. (a) PZT, (b) 4J29 Kovar alloy, (c) Au, (d) (Ti + Au), (e) (Ni + Au), (f) (Ti + Ni + Au); (g) Shear strength of different metallized PZT ceramics/SAC305 solder/4J29 Kovar alloy joints.

In this work, the thickness of individual metallization layers is listed in Table 1. Initially, a single layer of Au was employed to metallize the PZT ceramics to enhance the wetting ability due to its good affinity with solder [31]. However, the wetting state of solder on the single Au layer (Fig. 5(c)) was unsatisfactory as the Au dissolved into the solder led to direct contact with the PZT ceramics.

Table 1. The thickness of metallization layers.

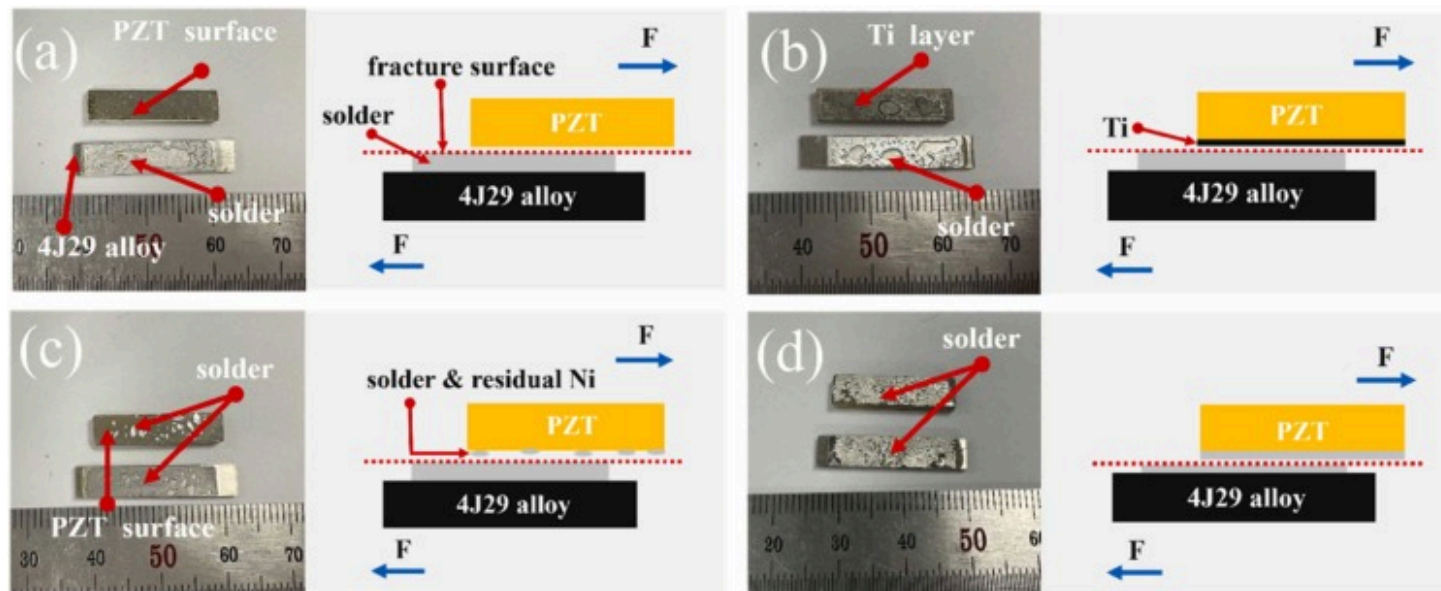
Metal layer	Thickness (nm)	Purity
Ti	150	99.99%

Metal layer	Thickness (nm)	Purity
Ni	1000	99.99%
Au	600	99.99%

Consequently, a double or multi metallization layers was necessary to enhance wetting properties of the solder on the PZT ceramics. Subsequently, (Ti + Au) was introduced as metallization layers on the PZT surface to evaluate solder's wetting properties as Ti has good chemical affinity with PZT. Nevertheless, as depicted in Fig. 5(d), the solder still could not wet the (Ti + Au) metallization layers due to the poor chemical compatibility between Ti and solder. The (Ni + Au) dual metallization layers were selected for metallizing PZT ceramics because Ni can serve as a barrier layer between the solder and PZT ceramics, given that the Ni-Sn binary system grows more slowly than the Au-Sn system. [32]. As shown in Fig. 5(e), the solder spread well on the (Ni + Au) as the result of the good wettability of solder on the metallization layers. Additionally, the (Ti + Ni + Au) multi-layer metallization approach was developed since Ti demonstrates superior bonding strength with PZT ceramics compared to Ni. As depicted in Fig. 5(f), the addition of Ti did not affect the solder's wettability on the (Ti + Ni + Au) metallization layers, with Ni acting as an effective barrier.

To further determine the optimal impact of the metallization layers, a test of joining strength was conducted. The metallized PZT ceramic and 4J29 Kovar alloy were joined using SAC305 solder. A shear strength test was carried out, with the results presented in Fig. 5(g). Shear strength of the joints was below 5.0 MPa when using single Au and (Ti + Au) as metallization layer. However, shear strength of the joints with (Ni + Au) as metallization layer increased to nearly 20.0 MPa. Furthermore, the addition of Ti layer between Ni and PZT ceramics could increase the shear strength to about 25.0 MPa. The fracture surfaces were analyzed, as shown in Fig. 6. The fracture surfaces of the PZT ceramics/Au/SAC305 solder/4J29 Kovar alloy joints were observed between the PZT ceramics and the solder, with no remaining solder on the surface of the PZT ceramics, as depicted in Fig. 6(a). This suggests that the interface between the PZT ceramics and the solder was the weakest link. The fracture surface of the PZT ceramics/ (Ti + Au)/SAC305 solder/4J29 Kovar alloy joints located between Ti layer and the solder, as shown in Fig. 6(b), showing that the bonding strength of Ti layer with the PZT ceramics was stronger than that of Ti layer with the solder. Similarly, no solder was found on the Ti layer. However, in the case of the PZT/ (Ni + Au)/SAC305 solder/4J29 Kovar alloy joints, as illustrated in Fig. 6(c), some solder particles remained on the surface of the PZT ceramics, suggesting that the bonding strength between the Ni layer and the solder was stronger than that between the solder and the Ti layer. Compared with the preceding three joints, the PZT

ceramics/ (Ti + Ni + Au)/SAC305 solder/4J29 Kovar alloy joints fractured at the solder, showing that bonding strength both solder/PZT ceramics and solder/4J29 Kovar alloy were higher than that of the solder itself. The addition of Ti offers a stable combination between metallization layers and the PZT ceramics due to Ti could react with O in the PZT ceramics to form Ti-O bond [33], and the generated oxide layer would match the lattice parameters and coefficients of thermal expand (CTE) with TiO_2 phase in the PZT ceramics [34], [35]. Based on good chemical compatibility and high shear strength, the (Ti + Ni + Au) metallization layer was preferred for the PZT ceramics/4J29 Kovar alloy joints.



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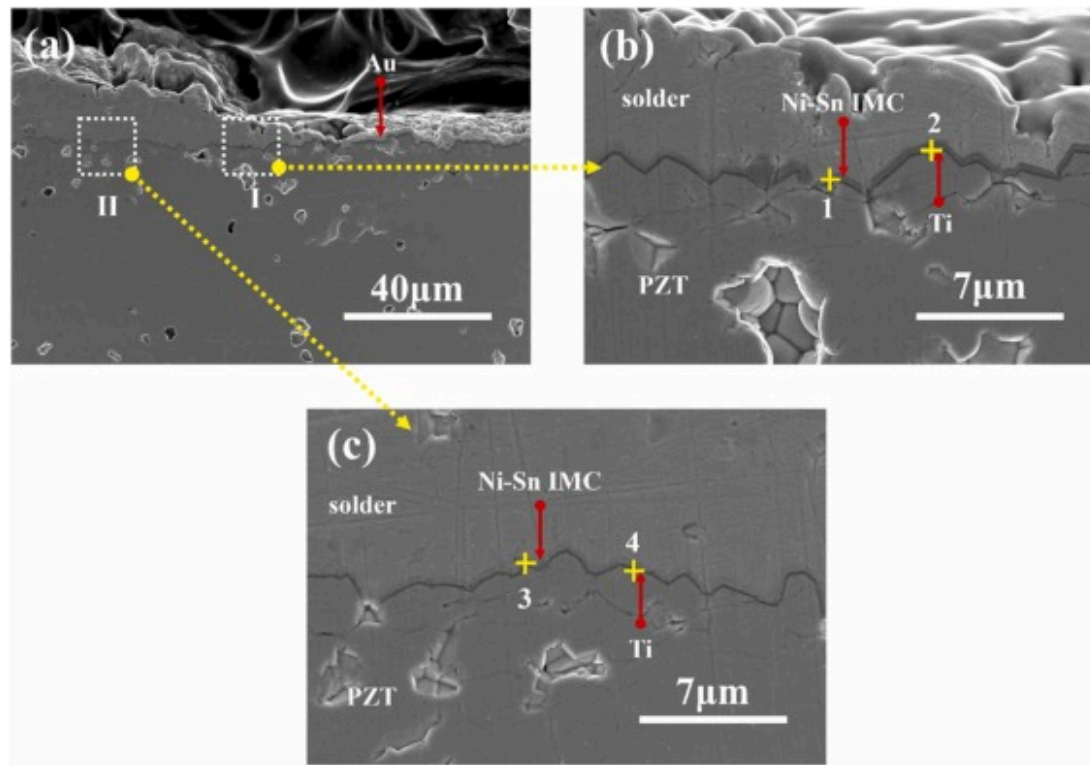
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Fig. 6. Shear fracture surface analysis of different metallization layers structures with surface layer of Au: (a)Au, (b)(Ti + Au), (c) (Ni + Au), (d)(Ti + Ni + Au).

3.2. Wetting behavior of SAC305 solder on the metallized PZT ceramics

Fig. 7(a) shows the SEM image: interface of SAC305 solder/ (Ti + Ni + Au)/PZT ceramics. The SAC305 solder exhibited excellent wettability on the (Ti + Ni + Au) metallization layers, which the contact angle is only about 13° . As shown in the Fig. 7, the Au layer was absolutely dissolved in the solder and the (Ti + Ni) layers were remained. According to EDS point scanning results of Figs.

7(b) and 7(c), listed in Table 2, point 1 was Ni rich area which contained the Ni and Ni-Sn intermetallic compounds (Ni-Sn IMCs). And point 3 had less Ni remained meant there was only almost Ni-Sn IMCs. The formation of different phases between points 1 and 3 can be attributed to the temperature increase, causing the solder to gradually melt, and spread over a larger area on the metallization layers. Consequently, this results in the central zone of the solder having a longer contact time compared to the surrounding zone. Point 3 was situated nearer to the central region of the solder, implying that the reaction time between Sn and Ni at point 3 was more extensive than at point 1, which was closer to the peripheral area. Similarly, point 4 was located closer to the central area of the solder compared to point 2, resulting in a greater diffusion of Sn through the Ni layer to interact with Ti at point 4 than at point 2.



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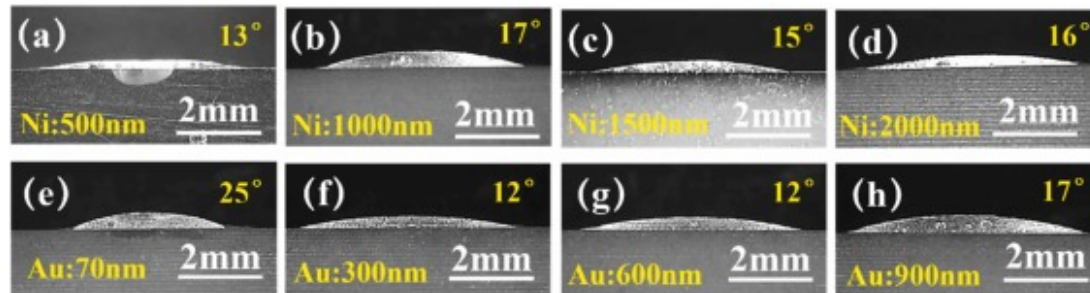
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Fig. 7. SEM image (a) wetting cross section of SAC305 solder on (Ti + Ni + Au); (b) the magnified image of the region I in (a); (c) the magnified image of the region II in (a).

Table 2. The EDS of each spot marked in Fig. 9(b) and (c).

Spot	Composition (at%)							Possible main phase
	O	Ti	Ni	Zr	Sn	Au	Pb	
1	16.61	5.72	67.62	1.49	4.14	1.98	2.43	Ni + Ni-Sn IMCs
2	28.49	35.11	25.74	3.19	2.04	-	5.43	Ti + Ni
3	33.04	5.14	29.23	3.07	25.35	1.71	2.46	Ni-Sn IMCs
4	39.65	36.81	0.89	4.42	10.13	-	8.10	Ti + β -Sn + PZT

Fig. 8(a)-(d) demonstrate how the thickness of the intermediate Ni layer affects the wetting angles, with a constant layering of 150 nm Ti and 600 nm Au. It was observed that variations in the thickness of the Ni layer exert minimal influence on the wetting angle. The impact of the surface layer Au on the wetting angles is presented in Fig. 8(e)-(h), where the layers of 150 nm Ti and 1000 nm Ni were kept constant. As the Au layer thickness increased, the wetting angle of the solder on the metallization layer first decreased, then increased. Except for the thinnest Au layer (70 nm), the solder all presented excellent wettability properties on the metallization layers.



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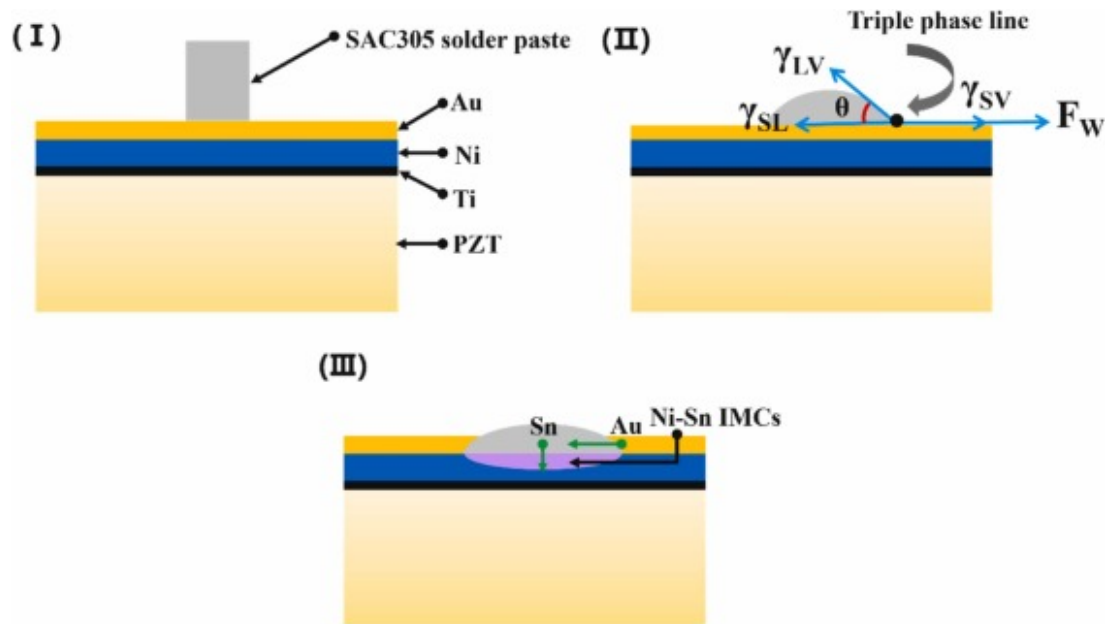
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Fig. 8. Effects of metallization layer thickness variation on wetting; (a)-(d): Ti:150 nm, Au:600 nm, the Ni thickness: 500 nm, 1000 nm, 1500 nm, and 2000 nm; (e)-(h): Ti:150 nm, Ni:1000 nm, the Au thickness: 70 nm, 300 nm, 600 nm, and 900 nm.

Thickness of the intermediate Ni layer had little influence on the wetting angle, indicating that wetting behavior was primarily governed by the Au layer. The wetting behavior of SAC305 solder on the metallization layers can be depicted as Fig. 9. The wetting driving force was determined by the following Eq. (2):

$$\gamma_{SV} + F_W - (\gamma_{SL} + \gamma_{LV} \cos \theta) = F_{WD} \quad (2)$$

Where the γ_{SV} is the interfacial tension between the metallization layers and atmosphere, the F_W is defined as the wetting force which was provided by the dissolution of Au into the solder, the γ_{SL} is the interfacial tension metallization layers in contact with molten solder, the γ_{LV} is the interfacial tension between the molten solder and atmosphere, θ is the contact angle, and the F_{WD} is the wetting driving force which facilitates the movement of triple phase line (interface of molten solder/metallization layers/atmosphere). At 255 °C, the solubility of Au in Sn can reach 12 wt% [36]. when the Au content kept within the solubility, Au can rapidly dissolve into Sn, facilitating the movement of the triple phase line. However, when the Au layer thickness was too thin (70 nm), Au quickly dissolved into the solder, and the triple phase line encountered Ni layer. Due to the low solubility of Ni in Sn (less than 1 wt%) [37], Ni rapidly reacted with Sn to form Ni-Sn IMCs, impeding the movement of the triple phase line as the increase of γ_{SL} [38]. On the other hand, when the Au layer became thick, large amount of Au dissolved into the solder and exceeded the solubility, AuSn₄ would form by the reaction of Au and Sn [39], also hindering the movement of the triple phase line. In summary, the thickness of the Au layer impacted the solder wettability, primarily due to the formation of IMC when the triple phase line moved. It is advantageous to control the thickness of the gold layer within a suitable range to enhance the solder wettability on the metallization layers. In addition, the longitudinal diffusion of the Sn element into Ni layer occurred after the lateral diffusion of the triple-phase line, which had nearly no effect on the wettability.



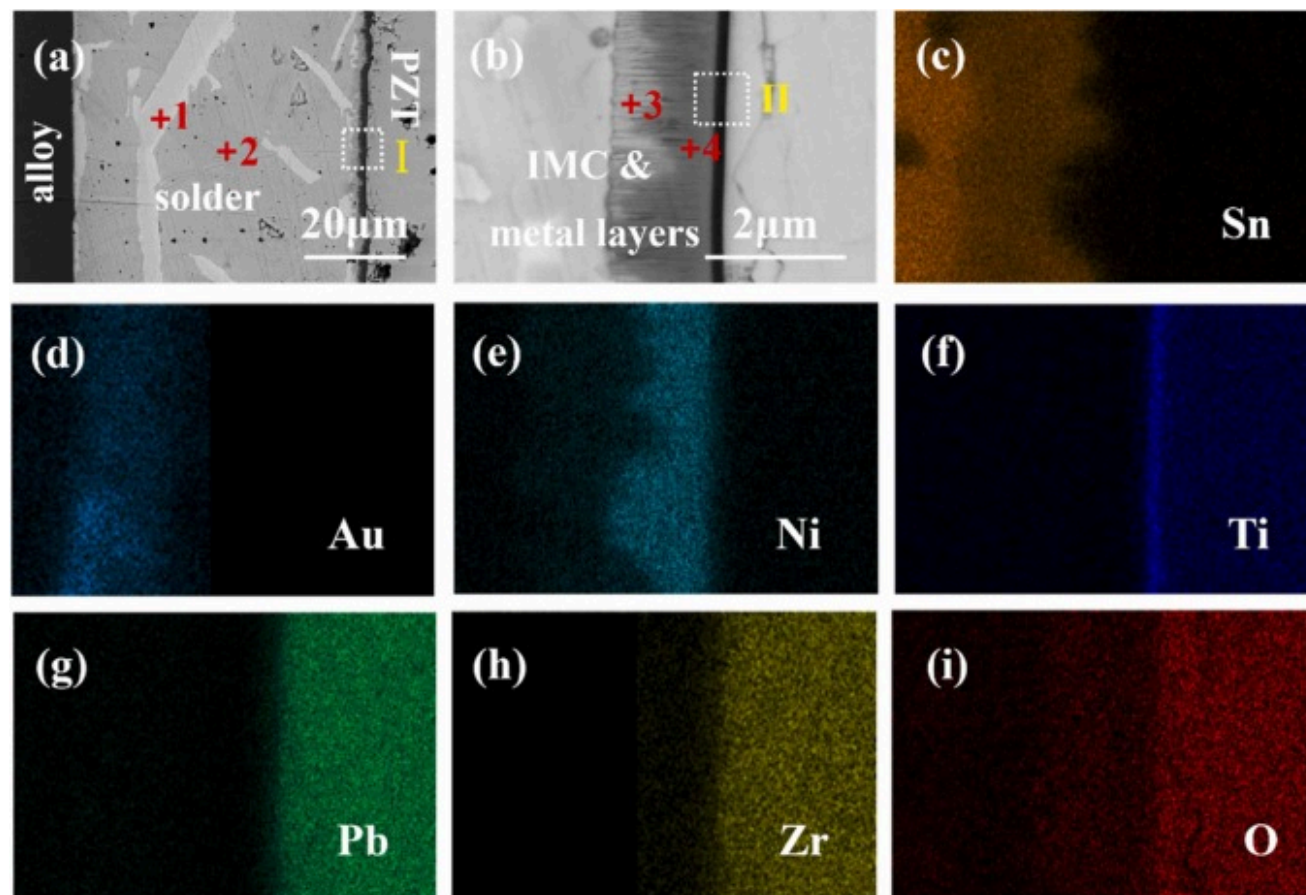
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Fig. 9. Wetting behavior of the SAC305 solder on the surface of (Ti + Ni + Au) metallization layers.

3.3. Microstructure and joining mechanism of the 4J29 Kovar alloy/SAC305 solder/PZT ceramics joints

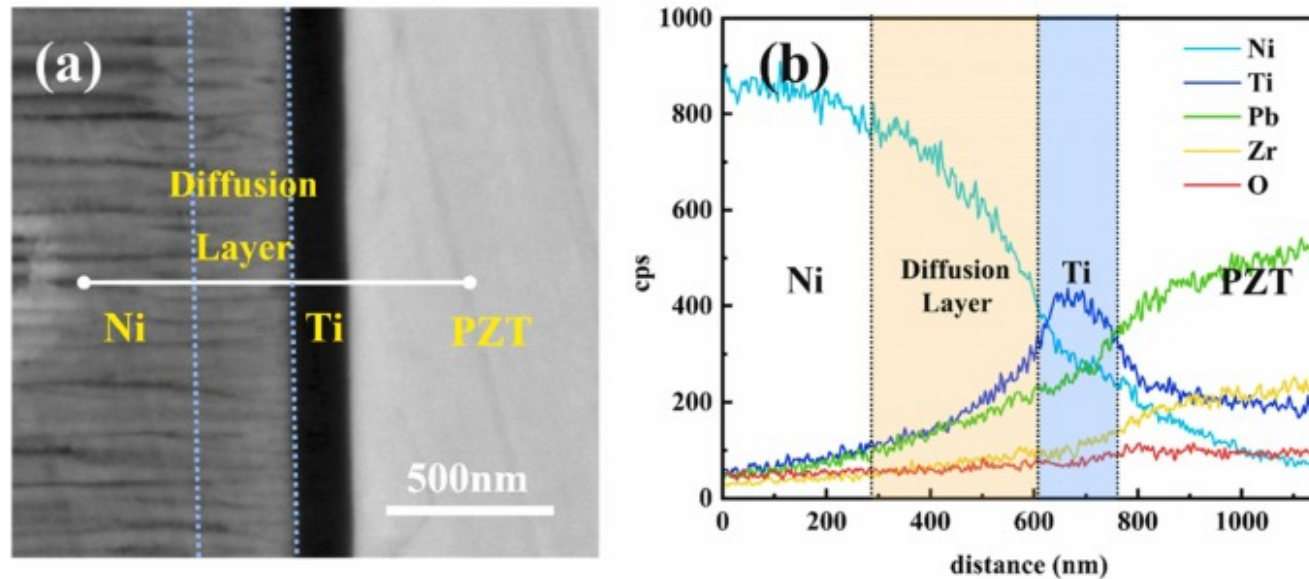
Fig. 10 shows the microstructure of 4J29 Kovar alloy/SAC305 solder/PZT ceramics joint when the metallization layers consisted of Ti (150 nm), Ni (1500 nm), Au (600 nm). As shown in Fig. 10(a) and (b), the solder bonded tightly with the 4J29 Kovar alloy and the metallized PZT ceramics, and no obvious defects were found. Fig. 10(c)-(i) illustrate the EDS mapping results of the joints. Element Au in the metallization layers completely dissolved into the solder, while Sn in the solder diffused into the Ni layer to form the Ni-Sn IMCs. Notably, the Ti layer in the metallization layers was only adjacent to the PZT ceramics, and no diffusion of Sn into the Ti layer was observed, thanks to the remaining Ni layer acting as a barrier. As depicted in Fig. 11(a) and (b), there is obvious mutual interdiffusion between Ni layer and Ti layer. The element Ni diffuses through the Ti layer and extends diffusively into the PZT to approximately 230 nm. Simultaneously, element Ti diffuses into the Ni layer, forming a diffusion layer with a thickness of approximately 260 nm. Furthermore, the tight bonding between Ti and Ni is achieved through the mutual interdiffusion of these two elements.



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Fig. 10. Microstructure of 4J29 Kovar alloy/SAC305 solder/metallized PZT ceramics joint after reflow soldering (a) the SEM of the joint; (b) the magnified image of the region I; (c)~(i): the corresponding EDS mapping results of the region I.



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Fig. 11. (a) The magnified image of the region II in Fig. 10; (b) the corresponding EDS line scanning of (a).

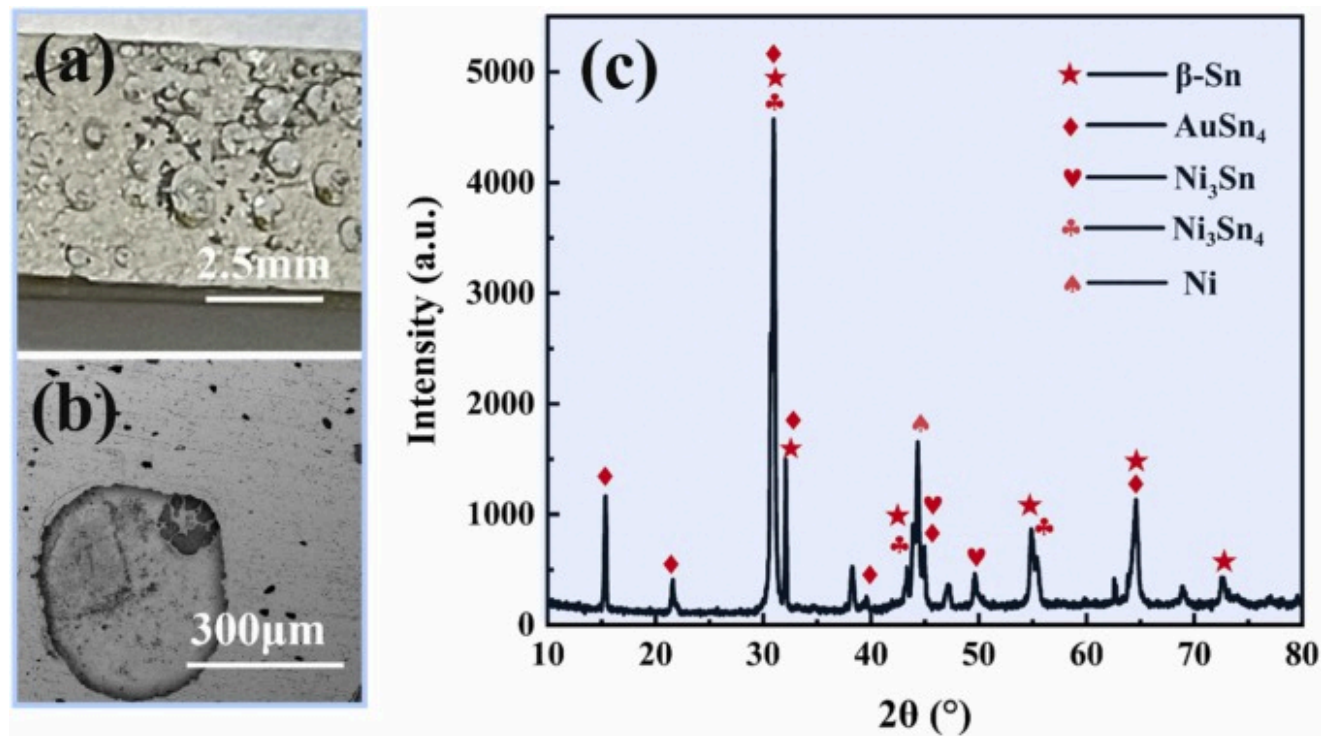
To identify the composition generated in the solder and metallization layers, the EDS point analysis was performed. Distinct phases within the solder region were identified and designated as point 1 and point 2, while the different phases within the Ni-Sn intermetallic compounds (IMCs) and metallization layers were labeled as points 3 and 4. The chemical compositions corresponding to points 1 through 4 are detailed in Table 3. Point 1 mainly contained Sn and Au, and since their atomic ratio was close to 4:1, it could be confirmed as AuSn_4 [39]. Point 2 only contained Sn, implying it was the substrate of the solder. Point 3 primarily consisted of element Sn and element Ni, inferring as Ni_3Sn_4 according to atomic ration. Point 4 mainly comprised Ni, indicating that it was residual Ni after Ni-Sn reaction. Fig. 12(a) and (b) depict the macroscopic and microscopic morphologies of the ceramic fracture surface, respectively. From the figures, it can be observed that the fracture surface is relatively smooth, and the fracture mainly occurs through the solder material. To further confirm the compositions in the joint, XRD analysis of the fracture joint of PZT ceramics was performed as shown in Fig.12(c). The β -Sn, AuSn_4 , Ni_3Sn_4 and Ni were found, which is consistent with the EDS. Furthermore, the Ni_3Sn was also found in the XRD. There are three Ni-Sn IMCs (Ni_3Sn_4 , Ni_3Sn_2 , Ni_3Sn)

[37] probably formed when Ni react with Sn. The following equations is about the Gibbs free energy at 255°C of the three IMCs:



Table3. The EDS of each spot marked in Fig. 11(a) and (b).

Spot	Composition (at%)						Possible phase
	Sn	Au	Ni	Ti	O	Cu	
1	82.23	17.77	-	-	-	-	AuSn ₄
2	100.00	-	-	-	-	-	β-Sn
3	41.42	1.09	31.55	none	21.06	4.88	Ni ₃ Sn ₄
4	1.24	-	87.54	0.70	10.52	-	Ni



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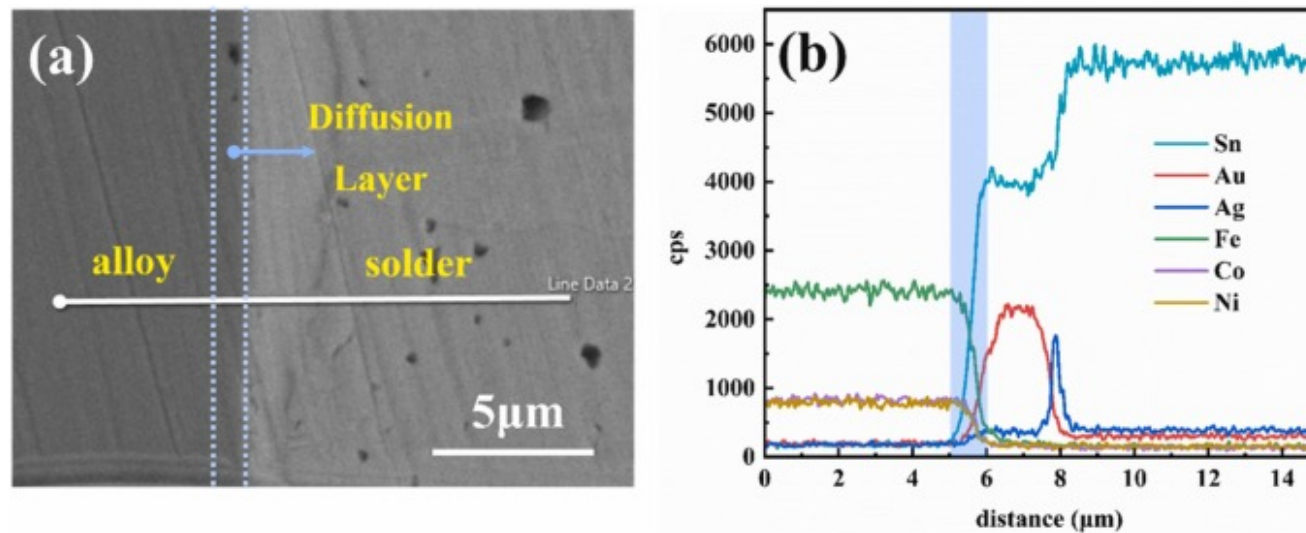
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Fig. 12. The fracture joint of PZT ceramics: (a) macro image; (b) SEM image; (c) XRD pattern.

The Ni₃Sn₄ phase is the most stable, attributed to its lowest Gibbs free energy; hence, the Ni-Sn intermetallic compounds (IMCs) predominantly consist of Ni₃Sn₄. Neither the EDS point analysis nor the XRD results indicated the presence of Ni₃Sn₂, suggesting that while it may form during the reaction process, it ultimately transforms completely into Ni₃Sn₄. Based on XRD analysis and the degree of Sn diffusion into Ni, there is a possibility that Ni₃Sn could persist in the region between Ni₃Sn₄ and Ni.

The formation of the Ni-Sn intermetallic compounds (IMCs) layer, stemming from the reaction between the solder and the ceramic metallization layers, could enhance the bond strength between the solder and metallization interfaces. However, due to the application of relatively low heating temperatures, a significant IMC layer was not observed at the interface between the 4J29 Kovar alloy and the solder. Fig. 13 presents the outcome of EDS line scanning across the 4J29 Kovar alloy/solder interface,

indicating that the adhesion between the alloy and the solder may primarily result from the diffusion of Sn from the solder into the alloy.



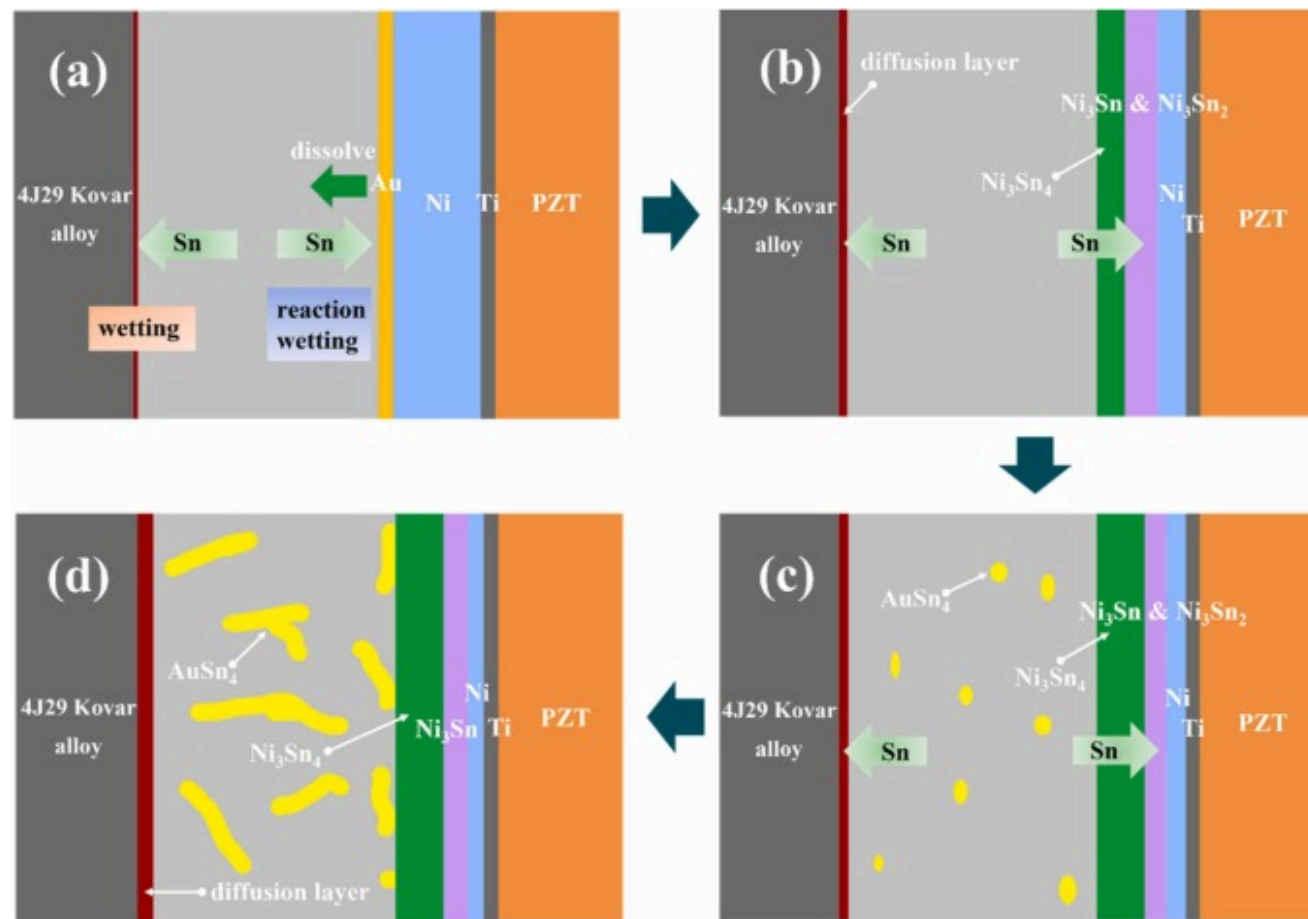
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Fig. 13. 4J29 Kovar alloy/SAC305 solder EDS line scan: (a) microstructure; (b) elemental distribution results.

According to microstructural analysis and metallurgical reaction theory, the joining mechanism of 4J29 Kovar alloy and metallized PZT can be summarized as the following four stages. In the initial stage, depicted as Fig.14(a), the liquid solder exhibited non-reactive wetting behavior on the 4J29 Kovar alloy side, where capillary forces facilitated uniform spreading of the liquid phase over the alloy surface and elements Sn diffused into the 4J29 Kovar alloy. Conversely, reactive wetting occurred on the metallized PZT ceramics side, dissolving of Au into Sn improved rapid spreading of solder over the surface of the metallization layers. In the second stage, depicted as Fig.14(b), element Sn continued to diffuse towards both sides. Once the Au in the metallization layers was fully dissolved into the solder, Sn commenced diffusion into the Ni layer, leading to the formation of Ni-Sn IMCs such as Ni_3Sn , Ni_3Sn_2 , and Ni_3Sn_4 , depending on the varying concentrations of Sn. In the third stage, when the temperature began to drop. Depicted as Fig.14(c), Au began to precipitate and reacted with Sn to form AuSn_4 due to decrease of solubility in Sn. In the last stage, as shown in Fig.14(d), as the temperature was further decreased, more Au precipitated and the

AuSn₄ phase grown up. In addition, the transformation of Ni₃Sn₂ to Ni₃Sn₄ occurred during the reflow soldering, driven by the attainment of a more stable state.



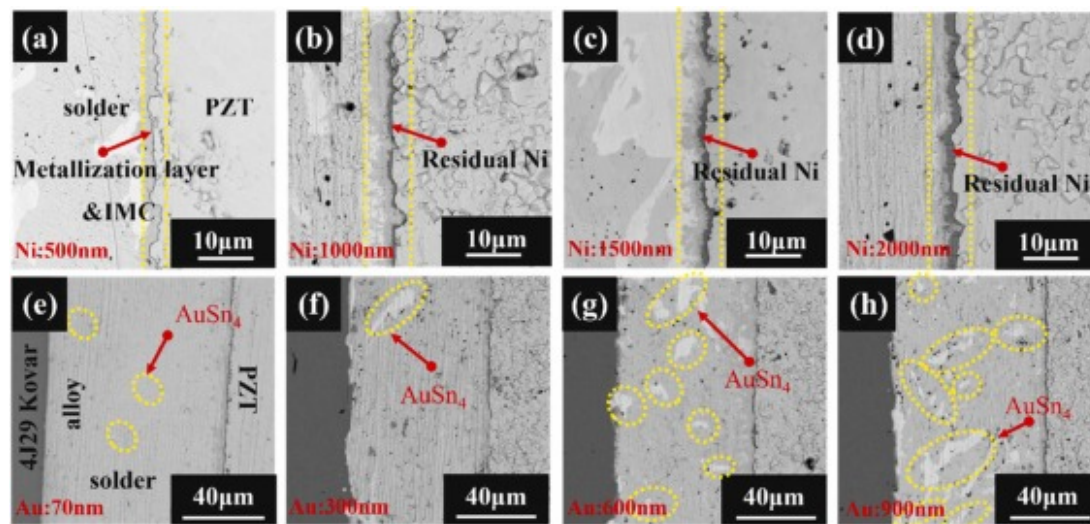
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Fig. 14. The schematic diagram of the microstructure evolution of the joint of 4J29 Kovar alloy/SAC305 solder/metallized PZT ceramics.

3.4. Optimization of the mechanical properties of the joints

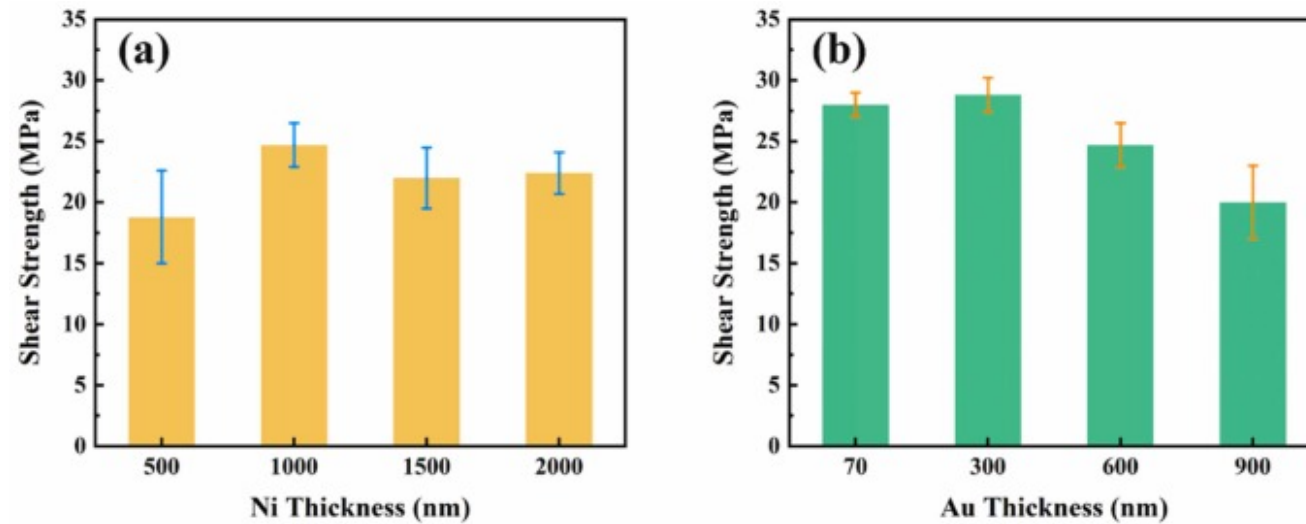
Fig.15(a)-(d) depict the influence of Ni layer thicknesses on the microstructure of the joints. When the thickness of Ni layer was 500 nm, Ni reacted completely with Sn, and no residual Ni was found in the metallization layers. As the thickness of Ni layer increased, the residual Ni layer obviously became thicker. Fig.16(a) illustrates the impact of Ni layer thicknesses on the strength of the joints. The shear strength of 18.8 ± 3.8 MPa, 24.7 ± 1.8 MPa, 22.0 ± 2.5 MPa, and 22.4 ± 1.7 MPa were obtained when the thicknesses of Ni layer were 500 nm, 1000 nm, 1500 nm, and 2000 nm, respectively. A notable increase of joints strength was observed when the Ni layer thickness increased from 500 nm to 1000 nm. However, with a further increase of the thickness of Ni layer, the change of joining strength became negligible. An inadequate thickness of the Ni layer resulted in the solder coming into direct contact with the Ti, leading to suboptimal wettability and the subsequent formation of voids. When the thickness Ni layer reached 1000 nm or above, residual Ni was found after the Ni-Sn reaction, acting as a barrier layer that impeded contact of the solder with Ti. Consequently, the optimal thickness of Ni in the metallization layers was determined to be 1000 nm.



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Fig. 15. The microstructure of the joint of different thickness of metallization layers: (a)-(d): Ti:150 nm, Au:600 nm, the Ni thickness changed to 500 nm, 1000 nm, 1500 nm, and 2000 nm; (e)-(h): Ti:150 nm, Ni:1000 nm, the Au thickness changed to 70 nm, 300 nm, 600 nm, and 900 nm.



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Fig. 16. Effect of metallization layers thickness on the shear strength of the joints: (a) Ni thickness; (b) Au thickness.

Fig.15(e)-(h) present the impact of the thicknesses of Au layer on the microstructure of the joints with a Ni layer thickness of 1000 nm. As the Au layer became thicker, the AuSn_4 content in the solder gradually increased. Fig.16(b) shows the effect of Au layer thicknesses on the strength of the joints with the fixed Ni layer thickness of 1000 nm. With an increase in the thickness of the gold (Au) layer, the shear strength demonstrates a pattern of initially rising before eventually declining. The shear strength of the joints was 28.0 ± 1.0 MPa for 70 nm, 28.8 ± 1.4 MPa for 300 nm, 24.7 ± 1.8 MPa for 600 nm, and 20.0 ± 3.0 MPa for 900 nm, respectively. With the thickness of Au layer increased, size of AuSn_4 phase became larger. As a brittle phase, small amount of AuSn_4 could improve the solder properties [39], however large amount of big-size AuSn_4 would lead to the decrease of the joining strength. So, the optimized Au layer thickness was 300 nm.

4. Conclusion

In this work, metallization layers system of (Ti + Ni + Au) was designed and prepared to metallize the PZT ceramics, and SAC305 solder was used to join the 4J29 Kovar alloy with the metallized PZT ceramics by reflow soldering process. The following

conclusions were drawn:

- (1) In comparison to Au and (Ti + Au) metallization layers, the combination of (Ni + Au) and (Ti + Ni + Au) metallization layers ensured good wettability of the SAC305 solder on the PZT ceramics. The bonding strength of (Ti + Ni + Au) with PZT ceramics was stronger (Ni + Au) due to the formation of Ti-O bonds with oxygen in PZT ceramics. Accordingly, for the PZT ceramics, the optimal metallization layers system was (Ti + Ni + Au).
- (2) The wetting behavior of SAC305 solder on the (Ti + Ni + Au) metallization layers was controlled by the dissolving of Au on the solder in the form of movement of the triple phase line. Insufficient Au content led to contact of the solder with Ni, resulting in the formation of Ni-Sn IMCs. Redundant Au also led to the generation of AuSn₄. The formation of IMCs impeded the lateral movement of the triple phase line, resulting in increase of wetting angles.
- (3) The microstructure of the 4J29 Kovar alloy/solder/metallized PZT ceramics joints with (Ti + Ni + Au) metallization layers was composed of 4J29 Kovar alloy/(solder+AuSn₄) +(Ni₃Sn₄+Ni₃Sn+Ni) +Ti/PZT ceramics. Sn reacted with Ni in the metallization layers to form intermetallic compounds, which was the main reason of the joining.
- (4) Thickness of the intermediate layer Ni and surface layer Au affected the microstructure and the shear strength of the joints. The highest shear strength of the joints reached 29.0±1.4 MPa which improved 120% compared with adhesive when thickness of the Ni layer was 1000 nm and thickness of the Au layer was 300 nm.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data Availability

Data will be made available on request.

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